

BeatUp Engine

HOW TO CHART

v1.1 · Silencer v1.9

A step-by-step guide from song to finished chart

OVERVIEW

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OVERVIEW

The Process at a Glance

1

Prepare the song

SILENCER V1.9

Silence a .wav of your song and keep the original as a backup.

2

Set the timing

SILENCER V1.9

Confirm BPM, offset and the Start Madi.

3

Chart

CHART EDITOR

Place notes on the grid: coarse first, then refine.

4

Review

AUTO-PLAY · SPLIT VIEW

Check timing and patterns, fix duplicates.

5

Export & test

PLAY YOUR CHART

Save the .bu, export .slk and play it.

Rule of thumb: Silencer solves the timing, the Editor builds the chart, and Play your chart proves that it works.

What You Need

Your song

FLAC WAV OGG

Silencing needs a .wav file.

The finished chart uses the silenced .ogg.

Keep the original audio as a backup.

BeatUp Engine v1.1

OR NEWER

Windows 10 / 11 (64-bit)

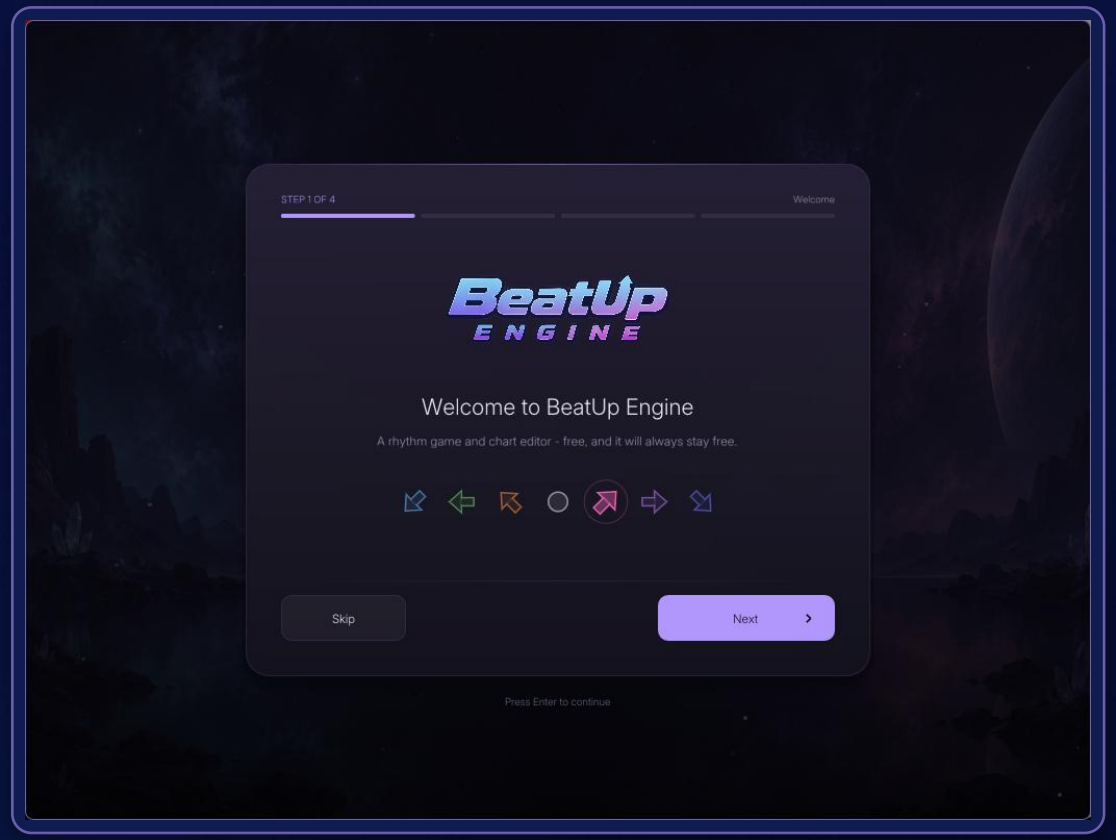
One self-contained BeatUpEngine.exe: graphics, fonts and sounds are built in.

Silencer v1.9 is included (Editor › Help › Open Silencer…).

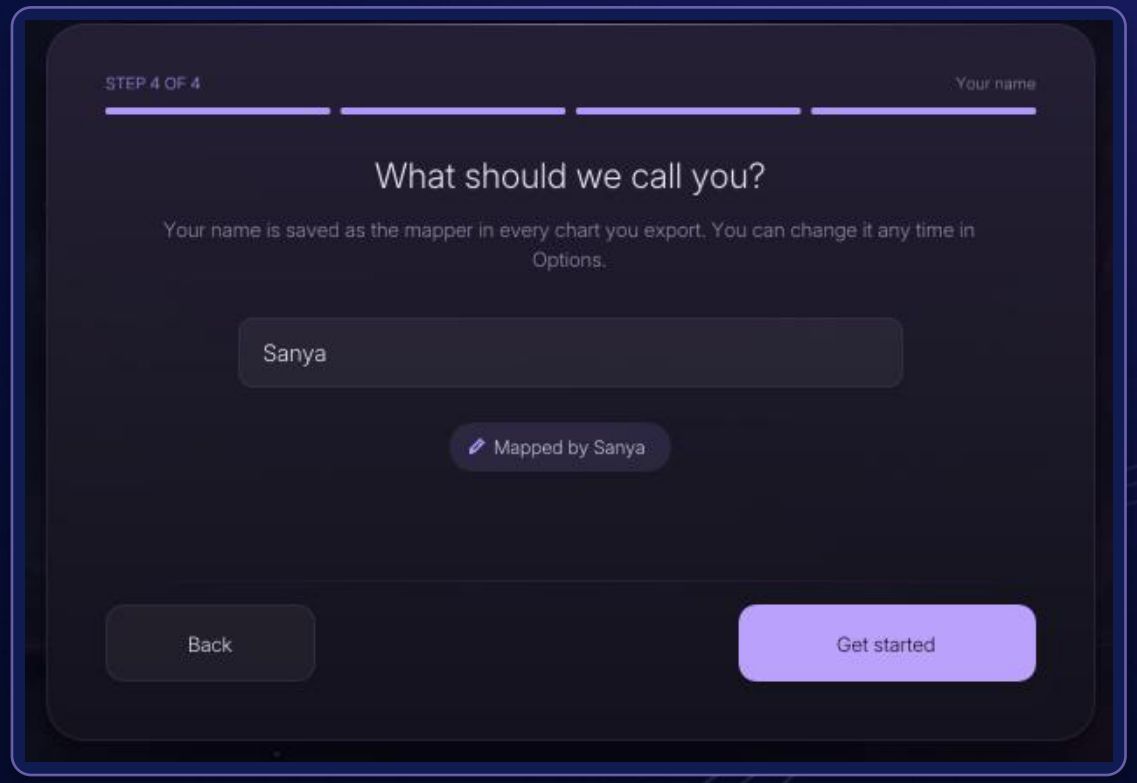
Folder structure

```
BeatUp Engine/
├ BeatUpEngine.exe
├ config.ini
├ charts/   your .bu files
├ assets/   optional
└ patcher/   optional
```

First Start



1 Onboarding takes four short steps. Press Next or Enter to continue.



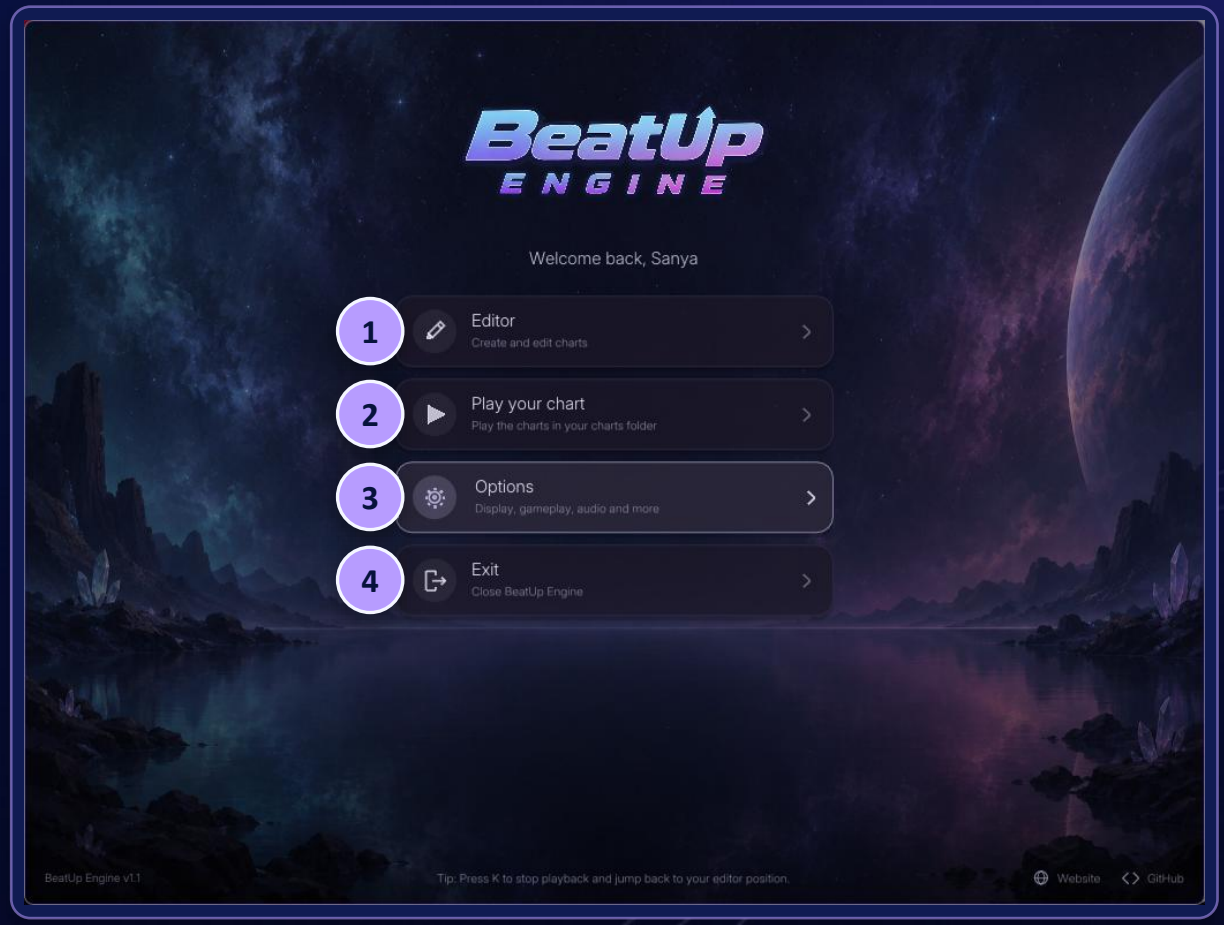
2 Enter your creditor name to watermark your charts for the future. You can change it later in Options › General.

Main Menu

The BeatUp Engine has four options on start-up.

1. **Editor**
   Create and edit charts
2. **Play your chart**
   Play the charts in your charts folder
3. **Options**
   Display, gameplay, audio and more
4. **Exit**
   Close BeatUp Engine

Custom assets: head to Options › General and turn on “Use custom assets”.



Terminology: BPM, Offset, Madi, Start Madi

Charters use certain terms to analyse music and find the correct timing.

BPM

The key value. The BPM you chart on has to match the song exactly. Advanced charters can manipulate the BPM, which is not part of this guide.

Offset

Relevant for every mode except Normal Mode. BeatUp charts should be made on a 0.7 offset; the lower the BPM, the higher the offset becomes.

Madi

The Korean word for a measure or bar. One Madi is one 4/4 bar.

Start Madi

The placement of your first note minus 1 bar (Madi). Usually in the range of 5 – 16.

Total Madi

The total number of bars in the song. It starts counting from 0.

How they fit together



Start Madi = bar of your first note − 1

Determining BPM & Offset

Getting BPM and offset right is the key to silencing a song.

The offset is different for every song, so it has to be measured each time. Two tools help you:

Always chart on the real BPM of the song. A wrong BPM can’t be fixed with the offset.

Need a tutorial? Click here.

BUILT IN

Silencer v1.9

Included in the BeatUp Engine. Open it from the Chart Editor via Help › Open Silencer…

EXTERNAL

Arrow Vortex

A popular external tool to find BPM and offset of a music file.

Silencing: Open Silencer

1 **Open the Chart Editor**

From the main menu choose Editor.

2 **Help › Open Silencer…**

A new window pops up.

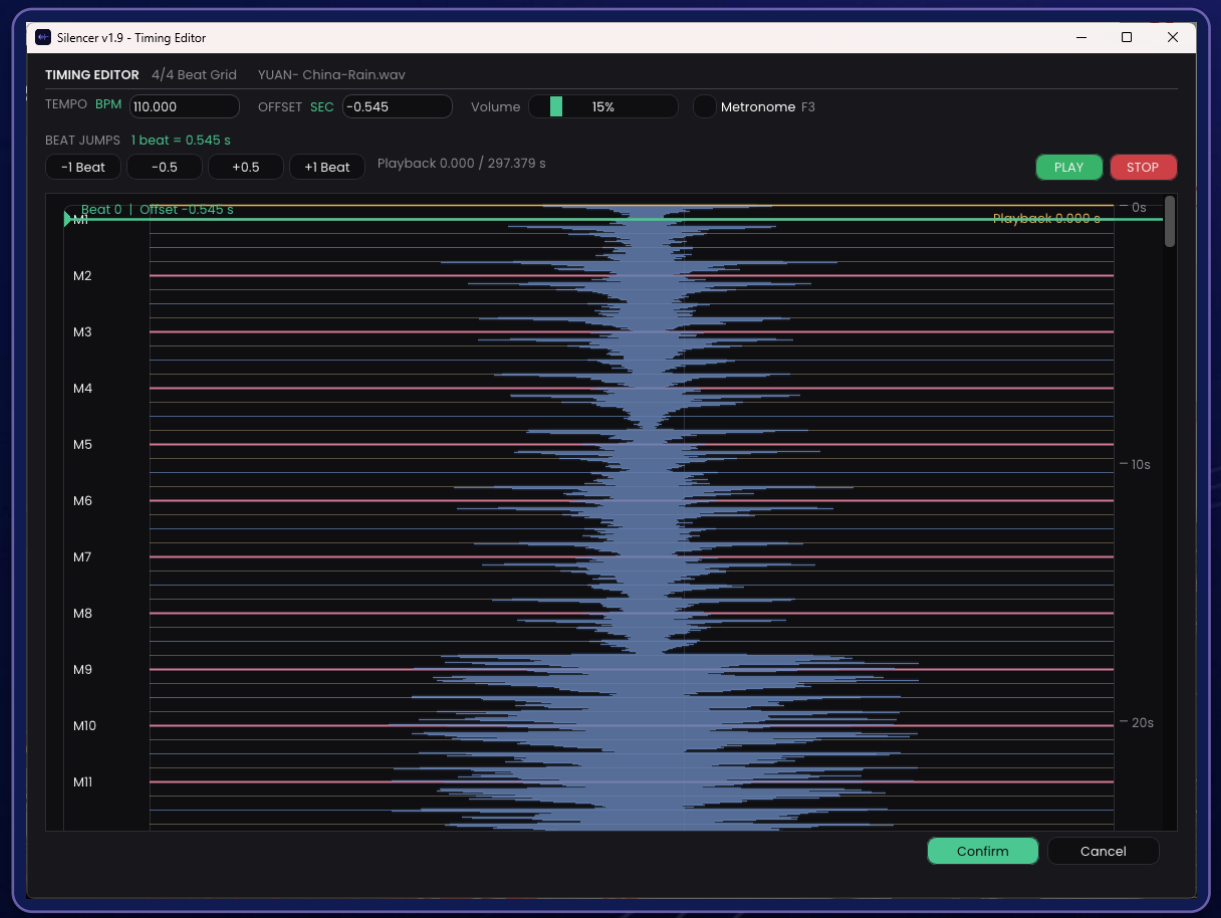
3 **Drop your song inside**

Important: to silence you have to drop a .wav file.

4 **Timing Editor opens**

Here you adjust the Offset & Tempo if you wish.

Editor › Help › Open Silencer…



Silencing: Set Timing

1 BPM

Auto-generated by Silencer. Check it against the song and correct it if needed.

2 Offset

Fine-tune with the Beat Jumps (−1, −0.5, +0.5, +1 beat). Turn on the metronome and press Play to check the grid by ear.

3 Start Madi

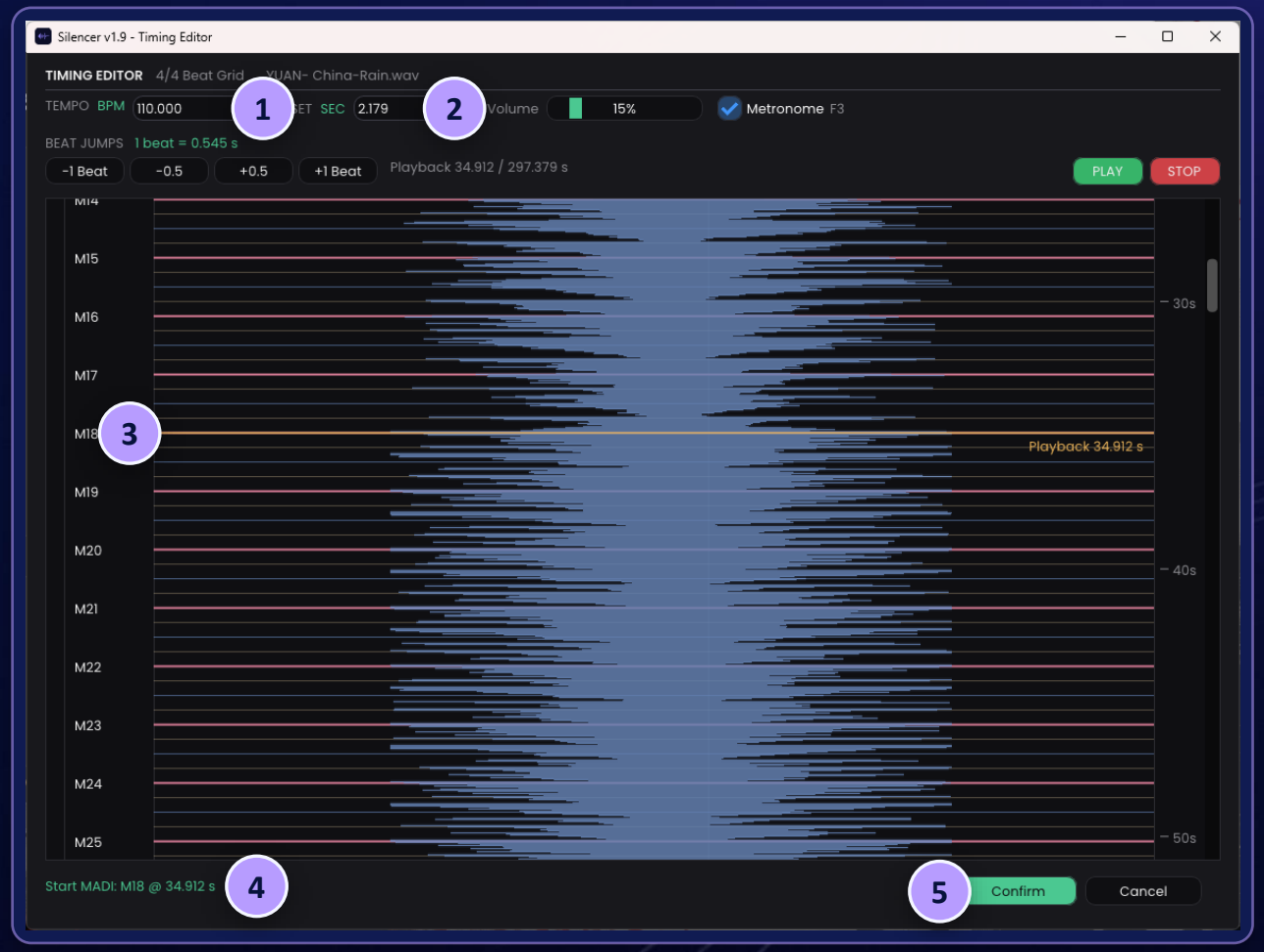
Click on the Madi where you want your Start Madi to be.

4 Status line

Shows that your Start Madi was selected successfully.

5 Confirm

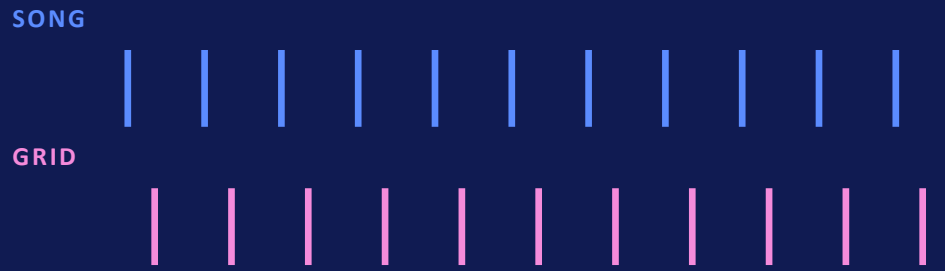
Applies your settings. Load the silenced song into the Editor (recommended: .ogg).



Drift Check

Is the song still in time further into the track? Jump to the end and listen.

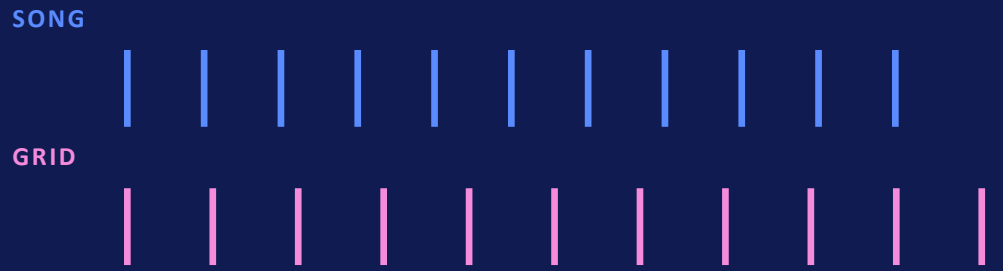
Everything is shifted by the same amount



Cause: Offset or Start Madi

Fix: Re-check the offset and the Start Madi in Silencer.

Right at the start, but it slowly drifts apart



Cause: BPM

Fix: Re-check the BPM. The offset sets the start; the BPM sets the slope.

Change one thing at a time. Never touch BPM, offset and Start Madi all at once.

Editor Basics

Opening the Editor gives you a blank preview. To load all assets, load your song file first.

Recommended: silenced .ogg

Don’t know how to silence? Click here.

File › New from Audio

or press Ctrl + Alt + A

The shortcuts follow a pattern:

Ctrl + Alt

means Open

Ctrl + Shift

means Save



Load Your Song

1 Pick your silenced song

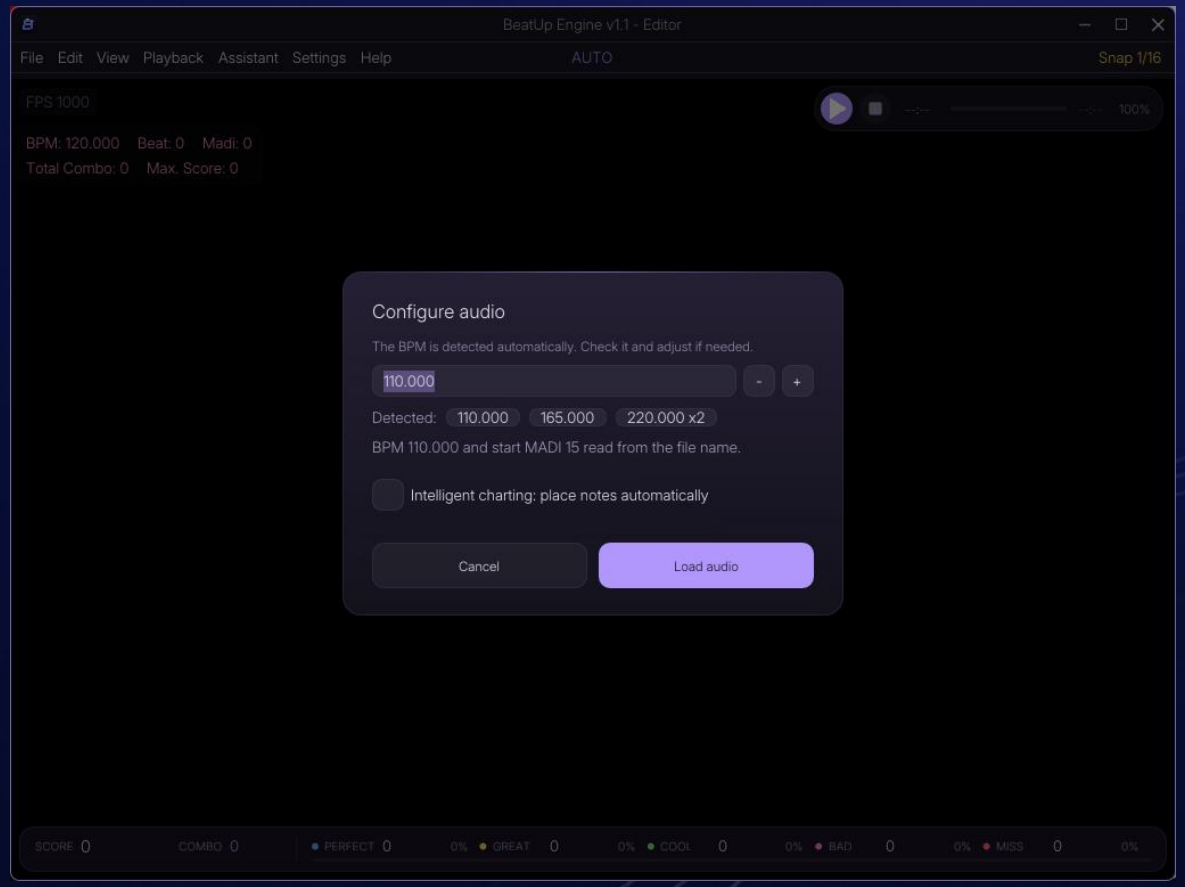
Choose the silenced .ogg you just created in Silencer.

2 Enter BPM and offset

Type in (or confirm) the BPM and offset you determined in Silencer.

3 Check the alignment

Press F4 for the metronome and play the first bars. Audio and grid have to match before you place many notes.



Save & Open Projects (.bu)

MySong.bu

ONE SELF-CONTAINED FILE

Your chart
all notes and timing

The song audio
embedded, no extra file

Bookmarks
your notes in the chart

Playback position
restored when you reopen

Save project

Ctrl + Shift + S

Saves the .bu with the audio embedded.

Open project

Ctrl + Alt + B

Or double-click any .bu file in Windows.

Share a chart

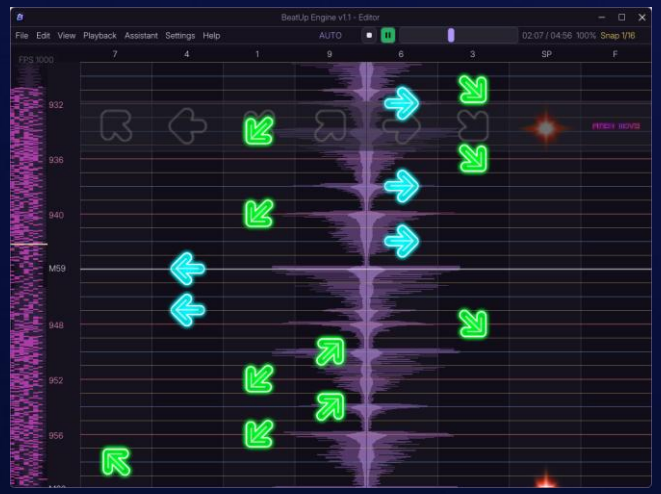
One .bu is all you need. Older .bu files without audio still open.

Editor Views



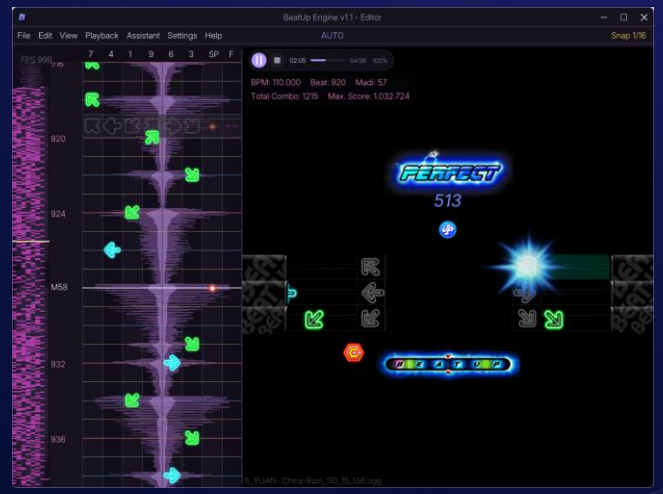
F1 Game view

The gameplay field, as players see it.



F2 Chart view

Editor only: the chart timeline.



F3 Split view

Gameplay and chart side by side.

Split view: drag the centre divider to resize both sides, double-click it to restore 50 / 50. The chart can run vertically or horizontally.

Columns & Keys

The editor uses the BeatUp column order. Six arrows, then Space and Finish Move.



7
numpad



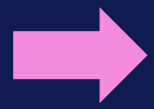
4
numpad



1
numpad



9
numpad



6
numpad



3
numpad

SPACE

SP
key S

FINISH

F
key F

Mouse

Left-click in a column to place or remove a note.

Numpad charting

When enabled, numpad 7 4 1 9 6 3 place the arrow notes.

Special keys

S places a Space note, F places a Finish Move.

The Snap Grid

ONE MADI (4 BEATS), SHOWN AT 1/16



Cycle the snap

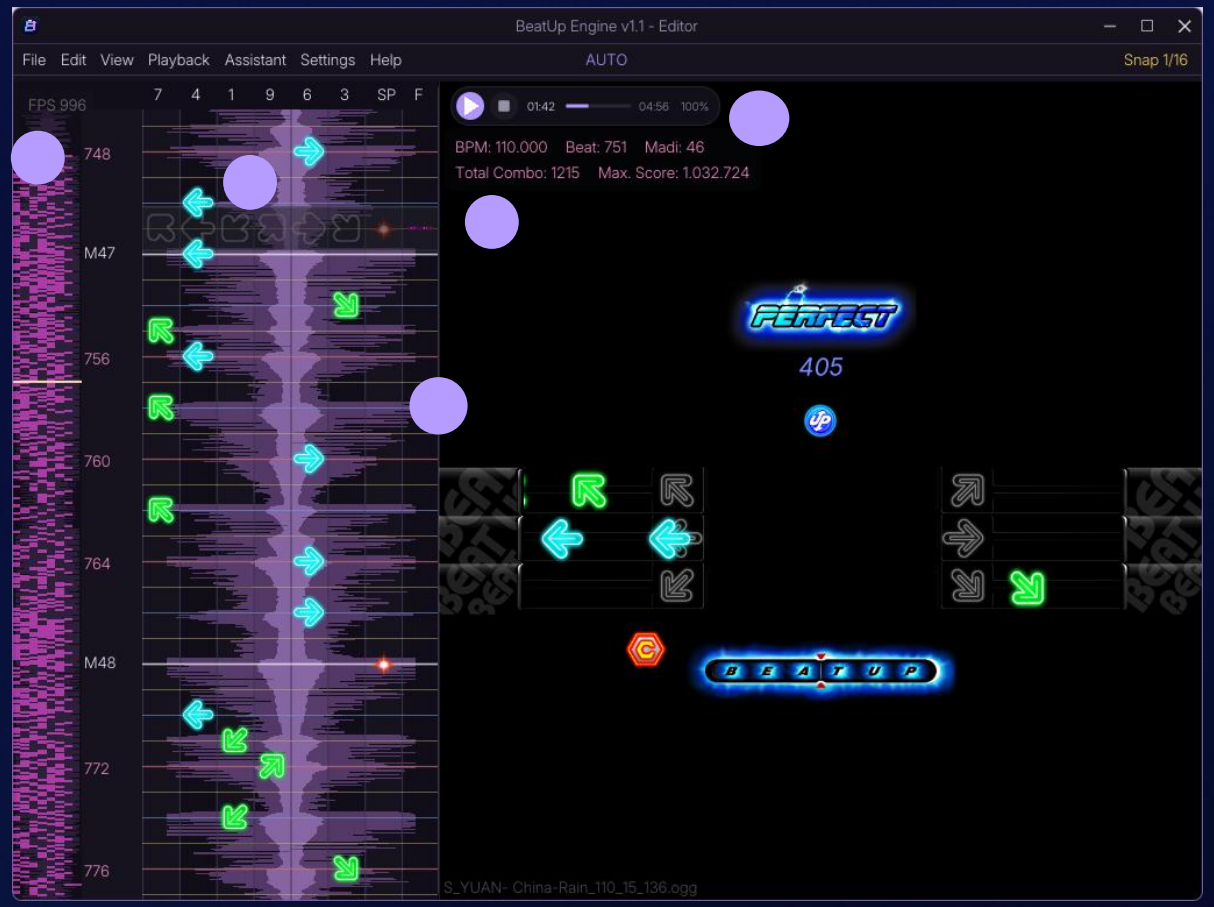
Alt + ← or Alt + →

Switches between 1/4, 1/8 and 1/16. The current snap is shown in the top-right corner of the editor.

Work coarse to fine

Build the skeleton on 1/4, then refine with 1/8 and 1/16.

Interface Tour



1. **Info card**
   BPM, Beat, Madi, Total Combo and Max. Score.

2. **Player**
   Play / pause, stop, time and playback speed.

3. **Waveform**
   Drawn behind the chart to help you find the beats.

4. **Receptor bar**
   Marks the hit position, just like in the game.

5. **Minimap**
   Shows where you are in the song.

6. **Score bar**
   Live score, combo and judgment counters.
   Found in Game Mode (F1).

Placing Notes & Navigation

Place notes

Left click	Place or remove a note
Right click	Open the editor context menu
S	Space note
F	Finish Move note
Numpad 7 4 1 9 6 3	Arrow notes (Numpad charting)

Navigate

↑ + ↓	Move by one beat
Shift + ↑ ↓	Move by one Madi
Mouse wheel	Scroll through the chart
Alt + ← →	Change the snap
F5	Play / pause (also Space)

Select, Copy, Paste & Mirror

Rectangle select

Ctrl + Drag

Draw a box around notes. The chart scrolls at the edges while you select.

Select all

Ctrl + A

Selects every note in the chart.

Copy & paste

Ctrl + C / Ctrl + V

Pasting replaces the notes in the destination, so sections copy 1:1.

Mirror

Ctrl + W / Ctrl + E

W mirrors the selection horizontally, E vertically.

Delete

Delete

Removes all selected notes.

Undo & redo

Ctrl + Z / Ctrl + Y

Every step is reversible and shows action feedback.

Space & Finish Move

SP

Space

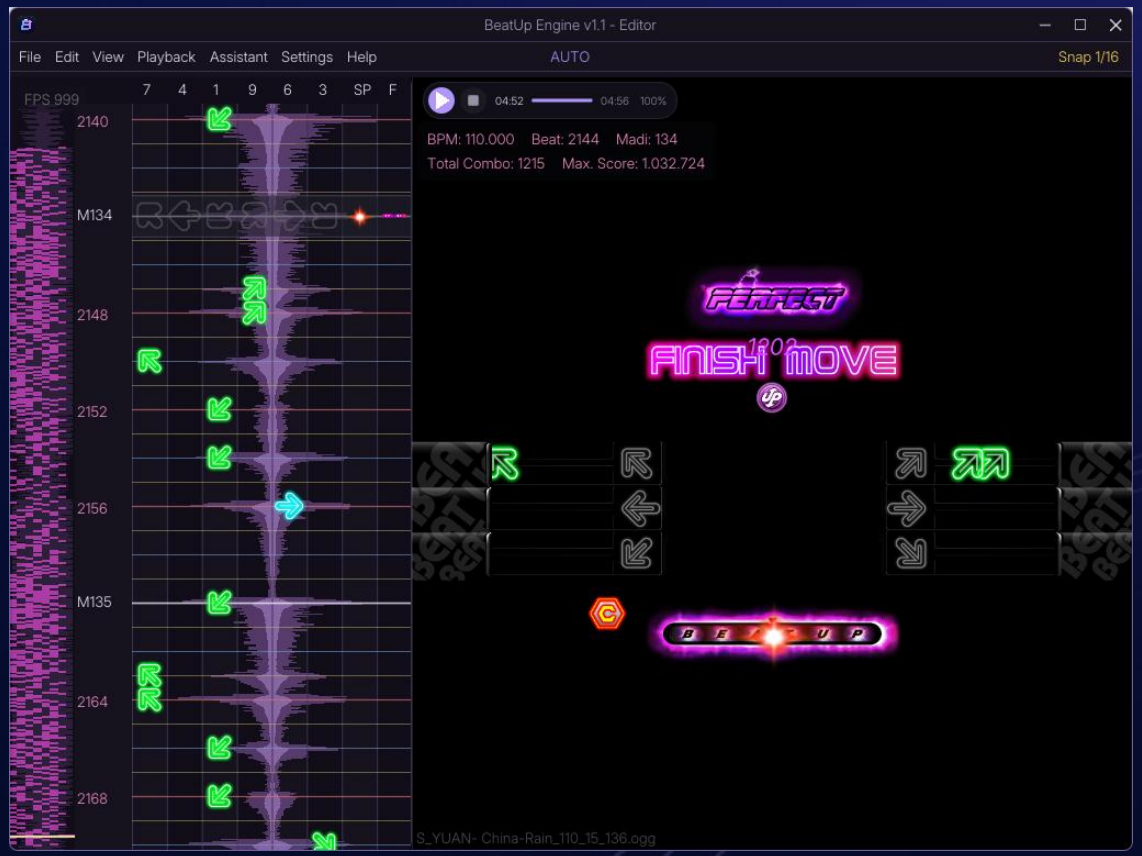
The Space column sits right of the six arrows. Place it with S or by clicking.

F

Finish Move

The last column is for Finish Move notes. Place it with F or by clicking.

Tip: check them in Game view (F1) with Auto-Play (F6). Some combinations on one beat count as duplicates, see Duplicate Check.



Charting Helpers

Auto-Play

F6

Plays your notes automatically, so you can judge timing and patterns.

Metronome

F4

Toggles the click track to check the grid by ear.

Playback speed

Ctrl + ← →

Changes the speed in 5% steps. Ctrl + 0 resets it to 100%.

Stop & return

F7 / K

F7 stops and resets. K stops and jumps back to your editor position.

Bookmarks

Ctrl + B

Add a note with your own text. Saved in the .bu: ideal when charting together.

Play / pause

F5 / Space

Starts and stops the audio from the current position.

What’s New in v1.0 – v1.1

Play your chart

v1.0.1

Play any .bu from the main menu, no Patcher needed.

Self-contained .bu

v1.0.1

Chart and music in one file, playback position restored.

Single-file .exe

v1.0.1

Everything is built in. Custom assets and backgrounds are optional.

Numpad charting

v1.0

Place arrow notes with the numeric keypad.

Snap hotkeys

v1.0

Alt + ← / → cycles 1/4, 1/8 and 1/16.

Project bookmarks

v1.0

Your own notes in the chart, saved in the .bu.

ALSO NEW

- One Options window
- BeatUp Night theme
- Skip Intro
- Theme Editor

Charting Tips

1/4 → 1/8 → 1/16

Start coarse, then refine.

1 Build the skeleton first

Place the main notes on 1/4, then add detail with 1/8 and 1/16.

2 Test the end of the song

Jump to the last chorus with Auto-Play on. Drift means the BPM is off.

3 Review in Split view

Press F3 and watch the game field while the chart plays.

4 Leave notes for yourself

Ctrl + B bookmarks mark spots to rework later.

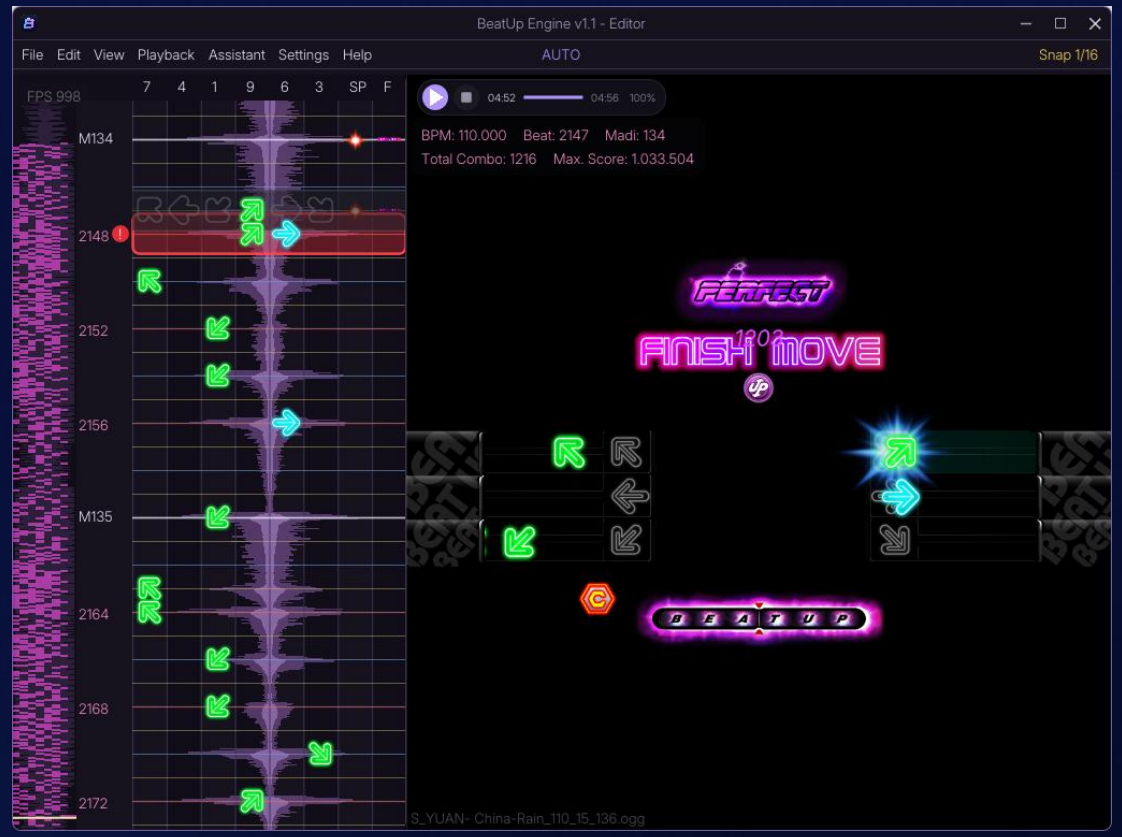
5 Save often

Ctrl + Shift + S writes the .bu, audio included.

6 Clear every red mark

Fix all duplicates before you export.

Duplicate Check



The editor detects duplicate notes live and marks them in red.

Allowed on the same beat

S + F **N + F**

N = arrow note, S = Space, F = Finish Move

Counts as a duplicate

All other same-beat conflicts, for example N + S.

Move or delete one of the notes until the red mark disappears.

Export

.slk

Chart export format

Export **Ctrl** + **Shift** + **E**

Import **Ctrl** + **Alt** + **E**

Your creditor name is written into every export.

.ssc

Chart with BPM and music filename

Export **Ctrl** + **Shift** + **V**

Import **Ctrl** + **Alt** + **V**

Carries the BPM and the name of the music file.

BEFORE YOU SHARE A CHART

1. Clear all red duplicates
2. Review with Auto-Play and Split view
3. Export the .slk (and .ssc)
4. Open the exported file again
5. Check audio, timing and credit

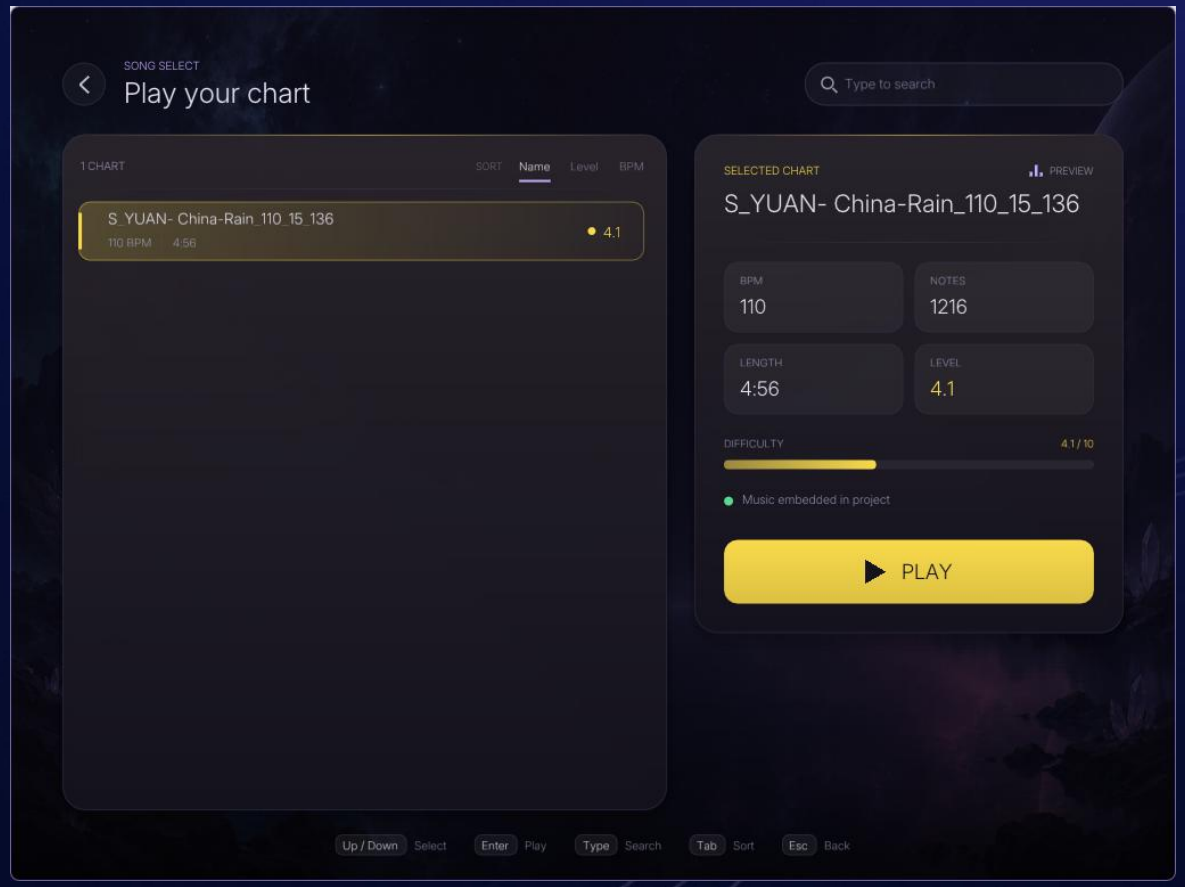
Play Your Chart

1. **Drop the .bu into charts/**
   Next to BeatUpEngine.exe. Sub-folders per song are fine.

2. **Main menu › Play your chart**
   Type to search, press Tab to sort by name, level or BPM.

3. **Pick a song**
   Stay on it for a short music preview and the detail card.

4. **Press Enter**
   Play the chart and test it.



The Patcher is optional: Options › General › Patcher.

APPENDIX

Troubleshooting

PROBLEM	WHAT TO CHECK
The chart looks shifted	Re-check the Start Madi and the offset in Silencer. Don’t hide a wrong BPM with a large offset.
Timing drifts later in the song	Re-check the BPM. The offset sets the start, the BPM sets the slope.
The song won’t silence	Silencer needs a .wav file. Convert your song first.
My chart is missing in Play your chart	The .bu has to be inside the charts/ folder next to BeatUpEngine.exe (sub-folders are fine).
An old project has no sound	Older .bu files without audio still open. Import the song again (Ctrl + Alt + A) and save.
My custom asset doesn’t show	Turn on Options › General › Use custom assets and check the folder and the file name.

APPENDIX · PRINT ME

Shortcut Cheat Sheet

PLAYBACK & VIEW	
F1 / F2 / F3	Game / Chart / Split
F4	Metronome
F5 / Space	Play / pause
F6	Auto-Play
F7	Stop & reset
K	Stop, back to editor
Alt + ← / →	Cycle snap
Ctrl + ← / →	Speed −5% / +5%
Ctrl + 0	Speed 100%
↑ / ↓	One beat
Shift + ↑ / ↓	One Madi

EDITING	
Left click	Place / remove note
Right click	Context menu
Ctrl + Drag	Rectangle select
Ctrl + A	Select all
Ctrl + C / V	Copy / paste
Ctrl + Z / Y	Undo / redo
Ctrl + W	Mirror horizontally
Ctrl + E	Mirror vertically
Ctrl + B	Bookmark
Delete	Delete selection
S / F	Space / Finish Move

PROJECT & APP	
Ctrl + Alt + A	Import audio
Ctrl + Alt + B	Open .bu
Ctrl + Shift + S	Save .bu
Ctrl + Alt + E	Import .slk
Ctrl + Shift + E	Export .slk
Ctrl + Alt + V	Import .ssc
Ctrl + Shift + V	Export .ssc
Ctrl + S	Options
Ctrl + Shift + R	Hard restart
F11	Fullscreen
Esc	Back / pause menu

APPENDIX

Happy Charting!

Questions, bugs or ideas? You will find everything here.

Website

web.beatupengine.workers.dev/bue

GitHub

github.com/bishpop/beatupengine

Releases

github.com/bishpop/beatupengine/releases